

3.C H-Spaces and Hopf Algebras

Of the three axioms for a group, it would seem that the least subtle is the existence of an identity element. However, we shall see in this section that when topology is added to the picture, the identity axiom becomes much more potent. To give a name to the objects we will be considering, define a space X to be an **H-space**, ‘H’ standing for ‘Hopf,’ if there is a continuous multiplication map $\mu: X \times X \rightarrow X$ and an ‘identity’ element $e \in X$ such that the two maps $X \rightarrow X$ given by $x \mapsto \mu(x, e)$ and $x \mapsto \mu(e, x)$ are homotopic to the identity through maps $(X, e) \rightarrow (X, e)$. In particular, this implies that $\mu(e, e) = e$.

In terms of generality, this definition represents something of a middle ground. One could weaken the definition by dropping the condition that the homotopies preserve the basepoint e , or one could strengthen it by requiring that e be a strict identity, without any homotopies. An exercise at the end of the section is to show the three possible definitions are equivalent if X is a CW complex. An advantage of allowing homotopies in the definition is that a space homotopy equivalent in the basepointed sense to an H-space is again an H-space. Imposing basepoint conditions is fairly standard in homotopy theory, and is usually not a serious restriction.

The most classical examples of H-spaces are **topological groups**, spaces X with a group structure such that both the multiplication map $X \times X \rightarrow X$ and the inversion map $X \rightarrow X$, $x \mapsto x^{-1}$, are continuous. For example, the group $GL_n(\mathbb{R})$ of invertible $n \times n$ matrices with real entries is a topological group when topologized as a subspace of the n^2 -dimensional vector space $M_n(\mathbb{R})$ of all $n \times n$ matrices over $\mathbb{R}$. It is an open subspace since the invertible matrices are those with nonzero determinant, and the determinant function $M_n(\mathbb{R}) \rightarrow \mathbb{R}$ is continuous. Matrix multiplication is certainly continuous, being defined by simple algebraic formulas, and it is not hard to see that matrix inversion is also continuous if one thinks for example of the classical adjoint formula for the inverse matrix.

Likewise $GL_n(\mathbb{C})$ is a topological group, as is the quaternionic analog $GL_n(\mathbb{H})$, though in the latter case one needs a somewhat different justification since determinants of quaternionic matrices do not have the good properties one would like. Since these groups GL_n over $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$ are open subsets of Euclidean spaces, they are examples of *Lie groups*, which can be defined as topological groups which are also manifolds. The GL_n groups are noncompact, being open subsets of Euclidean spaces, but they have the homotopy types of compact Lie groups $O(n)$, $U(n)$, and $Sp(n)$. This is explained in §3.D for $GL_n(\mathbb{R})$, and the other two cases are similar.

Among the simplest H-spaces from a topological viewpoint are the unit spheres S^1 in $\mathbb{C}$, S^3 in the quaternions $\mathbb{H}$, and S^7 in the octonions $\mathbb{O}$. These are H-spaces since the multiplications in these division algebras are continuous, being defined by

polynomial formulas, and are norm-preserving, $|ab| = |a||b|$, hence restrict to multiplications on the unit spheres, and the identity element of the division algebra lies in the unit sphere in each case. Both S^1 and S^3 are Lie groups since the multiplications in $\mathbb{C}$ and $\mathbb{H}$ are associative and inverses exist since $a\bar{a} = |a|^2 = 1$ if $|a| = 1$. However, S^7 is not a group since multiplication of octonions is not associative. Of course $S^0 = \{\pm 1\}$ is also a topological group, trivially. A famous theorem of J. F. Adams asserts that S^0 , S^1 , S^3 , and S^7 are the only spheres that are H-spaces; see §4.B for a fuller discussion.

Let us describe now some associative H-spaces where inverses fail to exist. Multiplication of polynomials provides an H-space structure on $\mathbb{CP}^\infty$ in the following way. A nonzero polynomial $a_0 + a_1z + \cdots + a_nz^n$ with coefficients $a_i \in \mathbb{C}$ corresponds to a point $(a_0, \dots, a_n, 0, \dots) \in \mathbb{C}^\infty - \{0\}$. Multiplication of two such polynomials determines a multiplication $\mathbb{C}^\infty - \{0\} \times \mathbb{C}^\infty - \{0\} \rightarrow \mathbb{C}^\infty - \{0\}$ which is associative, commutative, and has an identity element $(1, 0, \dots)$. Since $\mathbb{C}$ is commutative we can factor out by scalar multiplication by nonzero constants and get an induced product $\mathbb{CP}^\infty \times \mathbb{CP}^\infty \rightarrow \mathbb{CP}^\infty$ with the same properties. Thus $\mathbb{CP}^\infty$ is an associative, commutative H-space with a strict identity. Instead of factoring out by all nonzero scalars, we could factor out only by scalars of the form $\rho e^{2\pi i k/q}$ with ρ an arbitrary positive real, k an arbitrary integer, and q a fixed positive integer. The quotient of $\mathbb{C}^\infty - \{0\}$ under this identification, an infinite-dimensional lens space L^∞ with $\pi_1(L^\infty) \approx \mathbb{Z}_q$, is therefore also an associative, commutative H-space. This includes $\mathbb{RP}^\infty$ in particular.

The spaces $J(X)$ defined in §3.2 are also H-spaces, with the multiplication given by $(x_1, \dots, x_m)(y_1, \dots, y_n) = (x_1, \dots, x_m, y_1, \dots, y_n)$, which is associative and has an identity element (e) where e is the basepoint of X . One could describe $J(X)$ as the free associative H-space generated by X . There is also a commutative analog of $J(X)$ called the **infinite symmetric product** $SP(X)$ defined in the following way. Let $SP_n(X)$ be the quotient space of the n -fold product X^n obtained by identifying all n -tuples $(x_1, \dots, x_n)$ that differ only by a permutation of their coordinates. The inclusion $X^n \hookrightarrow X^{n+1}$, $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_n, e)$ induces an inclusion $SP_n(X) \hookrightarrow SP_{n+1}(X)$, and $SP(X)$ is defined to be the union of this increasing sequence of $SP_n(X)$'s, with the weak topology. Alternatively, $SP(X)$ is the quotient of $J(X)$ obtained by identifying points that differ only by permutation of coordinates. The H-space structure on $J(X)$ induces an H-space structure on $SP(X)$ which is commutative in addition to being associative and having a strict identity. The spaces $SP(X)$ are studied in more detail in §4.K.

The goal of this section will be to describe the extra structure which the multiplication in an H-space gives to its homology and cohomology. This is of particular interest since many of the most important spaces in algebraic topology turn out to be H-spaces.

Hopf Algebras

Let us look at cohomology first. Choosing a commutative ring R as coefficient ring, we can regard the cohomology ring $H^*(X; R)$ of a space X as an algebra over R rather than merely a ring. Suppose X is an H-space satisfying two conditions:

- (1) X is path-connected, hence $H^0(X; R) \approx R$.
- (2) $H^n(X; R)$ is a finitely generated free R -module for each n , so the cross product $H^*(X; R) \otimes_R H^*(X; R) \rightarrow H^*(X \times X; R)$ is an isomorphism.

The multiplication $\mu: X \times X \rightarrow X$ induces a map $\mu^*: H^*(X; R) \rightarrow H^*(X \times X; R)$, and when we combine this with the cross product isomorphism in (2) we get a map

$$H^*(X; R) \xrightarrow{\Delta} H^*(X; R) \otimes_R H^*(X; R)$$

which is an algebra homomorphism since both μ^* and the cross product isomorphism are algebra homomorphisms. The key property of Δ turns out to be that for any $\alpha \in H^n(X; R)$, $n > 0$, we have

$$\Delta(\alpha) = \alpha \otimes 1 + 1 \otimes \alpha + \sum_{0 < i < n} \alpha'_i \otimes \alpha''_{n-i} \quad \text{where } |\alpha'_j| = j = |\alpha''_j|$$

To verify this, let $i: X \rightarrow X \times X$ be the inclusion $x \mapsto (x, e)$ for e the identity element of X , and consider the commutative diagram

$$\begin{array}{ccccc} H^*(X; R) & \xrightarrow{\mu^*} & H^*(X \times X; R) & \xrightarrow{i^*} & H^*(X; R) \\ & \searrow \Delta & \times \uparrow \approx & \nearrow P & \times \uparrow \approx \\ & & H^*(X; R) \otimes_R H^*(X; R) & \xrightarrow{1 \otimes i^*} & H^*(X; R) \otimes_R H^*(e; R) \end{array}$$

The map P is defined by commutativity, and by looking at the lower right triangle we see that $P(\alpha \otimes 1) = \alpha$ and $P(\alpha \otimes \beta) = 0$ if $|\beta| > 0$. The H-space property says that $\mu i \approx 1$, so $P\Delta = 1$. This implies that the component of $\Delta(\alpha)$ in $H^n(X; R) \otimes_R H^0(X; R)$ is $\alpha \otimes 1$. A similar argument shows the component in $H^0(X; R) \otimes_R H^n(X; R)$ is $1 \otimes \alpha$.

We can summarize this situation by saying that $H^*(X; R)$ is a **Hopf algebra**, that is, a graded algebra $A = \bigoplus_{n \geq 0} A^n$ over a commutative base ring R , satisfying the following two conditions:

- (1) There is an identity element $1 \in A^0$ such that the map $R \rightarrow A^0$, $r \mapsto r \cdot 1$, is an isomorphism; one says A is *connected*.
- (2) There is a **diagonal** or **coproduct** $\Delta: A \rightarrow A \otimes A$, a homomorphism of graded algebras satisfying $\Delta(\alpha) = \alpha \otimes 1 + 1 \otimes \alpha + \sum_{0 < i < n} \alpha'_i \otimes \alpha''_{n-i}$ for $\alpha \in A^n$, $n > 0$, and $\alpha'_j, \alpha''_j \in A^j$.

Here and in what follows we take $\otimes$ to mean $\otimes_R$. The multiplication in $A \otimes A$ is given by the standard formula $(\alpha \otimes \beta)(\gamma \otimes \delta) = (-1)^{jk}(\alpha\gamma \otimes \beta\delta)$ where $\beta \in A^j$ and $\gamma \in A^k$. For a general Hopf algebra the multiplication is not assumed to be either associative or commutative (in the graded sense), though in the example of $H^*(X; R)$ for X an H-space the algebra structure is of course associative and commutative.

Example 3C.1. One of the simplest Hopf algebras is a polynomial ring $R[\alpha]$. The coproduct $\Delta(\alpha)$ must equal $\alpha \otimes 1 + 1 \otimes \alpha$ since the only elements of $R[\alpha]$ of lower dimension than α are the elements of R in dimension zero, so the terms α'_i and α''_{n-i} in the coproduct formula $\Delta(\alpha) = \alpha \otimes 1 + 1 \otimes \alpha + \sum_{0 < i < n} \alpha'_i \otimes \alpha''_{n-i}$ must be zero. The requirement that Δ be an algebra homomorphism then determines Δ completely. To describe Δ explicitly we distinguish two cases. If the dimension of α is even or if $2 = 0$ in R , then the multiplication in $R[\alpha] \otimes R[\alpha]$ is strictly commutative and $\Delta(\alpha^n) = (\alpha \otimes 1 + 1 \otimes \alpha)^n = \sum_i \binom{n}{i} \alpha^i \otimes \alpha^{n-i}$. In the opposite case that α is odd-dimensional, then $\Delta(\alpha^2) = (\alpha \otimes 1 + 1 \otimes \alpha)^2 = \alpha^2 \otimes 1 + 1 \otimes \alpha^2$ since $(\alpha \otimes 1)(1 \otimes \alpha) = \alpha \otimes \alpha$ and $(1 \otimes \alpha)(\alpha \otimes 1) = -\alpha \otimes \alpha$ if α has odd dimension. Thus if we set $\beta = \alpha^2$, then β is even-dimensional and we have $\Delta(\alpha^{2n}) = \Delta(\beta^n) = (\beta \otimes 1 + 1 \otimes \beta)^n = \sum_i \binom{n}{i} \beta^i \otimes \beta^{n-i}$ and $\Delta(\alpha^{2n+1}) = \Delta(\alpha\beta^n) = \Delta(\alpha)\Delta(\beta^n) = \sum_i \binom{n}{i} \alpha\beta^i \otimes \beta^{n-i} + \sum_i \binom{n}{i} \beta^i \otimes \alpha\beta^{n-i}$.

Example 3C.2. The exterior algebra $\Lambda_R[\alpha]$ on an odd-dimensional generator α is a Hopf algebra, with $\Delta(\alpha) = \alpha \otimes 1 + 1 \otimes \alpha$. To verify that Δ is an algebra homomorphism we must check that $\Delta(\alpha^2) = \Delta(\alpha)^2$, or in other words, since $\alpha^2 = 0$, we need to see that $\Delta(\alpha)^2 = 0$. As in the preceding example we have $\Delta(\alpha)^2 = (\alpha \otimes 1 + 1 \otimes \alpha)^2 = \alpha^2 \otimes 1 + 1 \otimes \alpha^2$, so $\Delta(\alpha)^2$ is indeed 0. Note that if α were even-dimensional we would instead have $\Delta(\alpha)^2 = \alpha^2 \otimes 1 + 2\alpha \otimes \alpha + 1 \otimes \alpha^2$, which would be 0 in $\Lambda_R[\alpha] \otimes \Lambda_R[\alpha]$ only if $2 = 0$ in R .

An element α of a Hopf algebra is called **primitive** if $\Delta(\alpha) = \alpha \otimes 1 + 1 \otimes \alpha$. As the preceding examples illustrate, if a Hopf algebra is generated as an algebra by primitive elements, then the coproduct Δ is uniquely determined by the product. This happens in a number of interesting special cases, but certainly not in general, as we shall see.

The existence of the coproduct in a Hopf algebra turns out to restrict the multiplicative structure considerably. Here is an important example illustrating this:

Example 3C.3. Suppose that the truncated polynomial algebra $F[\alpha]/(\alpha^n)$ over a field F is a Hopf algebra. Then α is primitive, just as it is in $F[\alpha]$, so if we assume either that α is even-dimensional or that F has characteristic 2, then the relation $\alpha^n = 0$ yields an equation

$$0 = \Delta(\alpha^n) = \alpha^n \otimes 1 + 1 \otimes \alpha^n + \sum_{0 < i < n} \binom{n}{i} \alpha^i \otimes \alpha^{n-i} = \sum_{0 < i < n} \binom{n}{i} \alpha^i \otimes \alpha^{n-i}$$

which implies that $\binom{n}{i} = 0$ in F for each i in the range $0 < i < n$. This is impossible if F has characteristic 0, and if the characteristic of F is $p > 0$ then it happens only when n is a power of p . For $p = 2$ this was shown in the proof of Theorem 3.20, and the argument given there works just as well for odd primes. Conversely, it is easy to check that if F has characteristic p then $F[\alpha]/(\alpha^{p^i})$ is a Hopf algebra, assuming still that α is even-dimensional if p is odd.

The characteristic 0 case of this result implies that $\mathbb{CP}^n$ is not an H-space for finite n , in contrast with $\mathbb{CP}^\infty$ which is an H-space as we saw earlier. Similarly, taking

$F = \mathbb{Z}_2$, we deduce that $\mathbb{RP}^n$ can be an H-space only if $n + 1$ is a power of 2. Indeed, $\mathbb{RP}^1 = S^1/\pm 1$, $\mathbb{RP}^3 = S^3/\pm 1$, and $\mathbb{RP}^7 = S^7/\pm 1$ have quotient H-space structures from S^1 , S^3 and S^7 since -1 commutes with all elements of S^1 , S^3 , or S^7 . However, these are the only cases when $\mathbb{RP}^n$ is an H-space since, by an exercise at the end of this section, the universal cover of an H-space is an H-space, and S^1 , S^3 , and S^7 are the only spheres that are H-spaces, by the theorem of Adams mentioned earlier.

It is an easy exercise to check that the tensor product of Hopf algebras is again a Hopf algebra, with the coproduct $\Delta(\alpha \otimes \beta) = \Delta(\alpha) \otimes \Delta(\beta)$. So the preceding examples yield many other Hopf algebras, tensor products of polynomial, truncated polynomial, and exterior algebras on any number of generators. The following theorem of Hopf is a partial converse:

Theorem 3C.4. *If A is a commutative, associative Hopf algebra over a field F of characteristic 0, and A^n is finite-dimensional over F for each n , then A is isomorphic as an algebra to the tensor product of an exterior algebra on odd-dimensional generators and a polynomial algebra on even-dimensional generators.*

There is an analogous theorem of Borel when F is a finite field of characteristic p . In this case A is again isomorphic to a tensor product of single-generator Hopf algebras, of one of the following types:

- $F[\alpha]$, with α even-dimensional if $p \neq 2$.
- $\Lambda_F[\alpha]$ with α odd-dimensional.
- $F[\alpha]/(\alpha^{p^i})$, with α even-dimensional if $p \neq 2$.

For a proof see [Borel 1953] or [Kane 1988].

Proof of 3C.4: Since A^n is finitely generated over F for each n , we may choose algebra generators $x_1, x_2, \dots$ for A with $x_i \in A^{|x_i|}$ and $|x_i| \leq |x_{i+1}|$ for all i . Let A_n be the subalgebra generated by $x_1, \dots, x_n$. This is a Hopf subalgebra of A , that is, $\Delta(A_n) \subset A_n \otimes A_n$, since $\Delta(x_i)$ involves only x_i and terms of smaller dimension. We may assume x_n does not lie in A_{n-1} . Since A is associative and commutative, there is a natural surjection $A_{n-1} \otimes F[x_n] \rightarrow A_n$ if $|x_n|$ is even, or $A_{n-1} \otimes \Lambda_F[x_n] \rightarrow A_n$ if $|x_n|$ is odd. By induction on n it will suffice to prove these surjections are injective. Thus in the two cases we must rule out nontrivial relations $\sum_i \alpha_i x_n^i = 0$ and $\alpha_0 + \alpha_1 x_n = 0$, respectively, with coefficients $\alpha_i \in A_{n-1}$.

Let I be the ideal in A_n generated by x_n^2 and the positive-dimensional elements of A_{n-1} , so I consists of the polynomials $\sum_i \alpha_i x_n^i$ with coefficients $\alpha_i \in A_{n-1}$, the first two coefficients α_0 and α_1 having trivial components in A^0 . Note that $x_n \notin I$ since elements of I having dimension $|x_n|$ must lie in A_{n-1} . Consider the composition

$$A_n \xrightarrow{\Delta} A_n \otimes A_n \xrightarrow{q} A_n \otimes (A_n/I)$$

with q the natural quotient map. By the definition of I , this composition $q\Delta$ sends $\alpha \in A_{n-1}$ to $\alpha \otimes 1$ and x_n to $x_n \otimes 1 + 1 \otimes \bar{x}_n$ where $\bar{x}_n$ is the image of x_n in A_n/I .

In case $|x_n|$ is even, applying $q\Delta$ to a nontrivial relation $\sum_i \alpha_i x_n^i = 0$ gives

$$0 = \sum_i (\alpha_i \otimes 1)(x_n \otimes 1 + 1 \otimes \bar{x}_n)^i = (\sum_i \alpha_i x_n^i) \otimes 1 + \sum_i i \alpha_i x_n^{i-1} \otimes \bar{x}_n$$

Since $\sum_i \alpha_i x_n^i = 0$, this implies that $\sum_i i \alpha_i x_n^{i-1} \otimes \bar{x}_n$ is zero in the tensor product $A_n \otimes (A_n/I)$, hence $\sum_i i \alpha_i x_n^{i-1} = 0$ since $x_n \notin I$ implies $\bar{x}_n \neq 0$. The relation $\sum_i i \alpha_i x_n^{i-1} = 0$ has lower degree than the original relation, and is not the trivial relation since F has characteristic 0, $\alpha_i \neq 0$ implying $i \alpha_i \neq 0$ if $i > 0$. Since we could assume the original relation had minimum degree, we have reached a contradiction.

The case $|x_n|$ odd is similar. Applying $q\Delta$ to a relation $\alpha_0 + \alpha_1 x_n = 0$ gives $0 = \alpha_0 \otimes 1 + (\alpha_1 \otimes 1)(x_n \otimes 1 + 1 \otimes \bar{x}_n) = (\alpha_0 + \alpha_1 x_n) \otimes 1 + \alpha_1 \otimes \bar{x}_n$. Since $\alpha_0 + \alpha_1 x_n = 0$, we get $\alpha_1 \otimes \bar{x}_n = 0$, which implies $\alpha_1 = 0$ and hence $\alpha_0 = 0$. $\square$

The structure of Hopf algebras over $\mathbb{Z}$ is much more complicated than over a field. Here is an example that is still fairly simple.

Example 3C.5: Divided Polynomial Algebras. We showed in Proposition 3.22 that the H-space $J(S^n)$ for n even has $H^*(J(S^n); \mathbb{Z})$ a divided polynomial algebra, the algebra $\Gamma_{\mathbb{Z}}[\alpha]$ with additive generators α_i in dimension ni and multiplication given by $\alpha_1^k = k! \alpha_k$, hence $\alpha_i \alpha_j = \binom{i+j}{i} \alpha_{i+j}$. The coproduct in $\Gamma_{\mathbb{Z}}[\alpha]$ is uniquely determined by the multiplicative structure since $\Delta(\alpha_1^k) = (\alpha_1 \otimes 1 + 1 \otimes \alpha_1)^k = \sum_i \binom{k}{i} \alpha_1^i \otimes \alpha_1^{k-i}$, which implies that $\Delta(\alpha_1^k/k!) = \sum_i (\alpha_1^i/i!) \otimes (\alpha_1^{k-i}/(k-i!))$, that is, $\Delta(\alpha_k) = \sum_i \alpha_i \otimes \alpha_{k-i}$. So in this case the coproduct has a simpler description than the product.

It is interesting to see what happens to the divided polynomial algebra $\Gamma_{\mathbb{Z}}[\alpha]$ when we change to field coefficients. Clearly $\Gamma_{\mathbb{Q}}[\alpha]$ is the same as $\mathbb{Q}[\alpha]$. In contrast with this, $\Gamma_{\mathbb{Z}_p}[\alpha]$, with multiplication defined by $\alpha_i \alpha_j = \binom{i+j}{i} \alpha_{i+j}$, happens to be isomorphic as an algebra to the infinite tensor product $\bigotimes_{i \geq 0} \mathbb{Z}_p[\alpha_{p^i}]/(\alpha_{p^i}^p)$, as we will show in a moment. However, as Hopf algebras these two objects are different since α_{p^i} is primitive in $\bigotimes_{i \geq 0} \mathbb{Z}_p[\alpha_{p^i}]/(\alpha_{p^i}^p)$ but not in $\Gamma_{\mathbb{Z}_p}[\alpha]$ when $i > 0$, since the coproduct in $\Gamma_{\mathbb{Z}_p}[\alpha]$ is given by $\Delta(\alpha_k) = \sum_i \alpha_i \otimes \alpha_{k-i}$.

Now let us show that there is an algebra isomorphism

$$\Gamma_{\mathbb{Z}_p}[\alpha] \approx \bigotimes_{i \geq 0} \mathbb{Z}_p[\alpha_{p^i}]/(\alpha_{p^i}^p)$$

Since $\Gamma_{\mathbb{Z}_p}[\alpha] = \Gamma_{\mathbb{Z}}[\alpha] \otimes \mathbb{Z}_p$, this is equivalent to:

(*) The element $\alpha_1^{n_0} \alpha_p^{n_1} \cdots \alpha_{p^k}^{n_k}$ in $\Gamma_{\mathbb{Z}}[\alpha]$ is divisible by p iff $n_i \geq p$ for some i .

The product $\alpha_1^{n_0} \alpha_p^{n_1} \cdots \alpha_{p^k}^{n_k}$ equals $m \alpha_n$ for $n = n_0 + n_1 p + \cdots + n_k p^k$ and some integer m . The question is whether p divides m . We will show:

(**) $\alpha_n \alpha_{p^k}$ is divisible by p iff $n_k = p - 1$, assuming that $n_i < p$ for each i .

This implies (*) by an inductive argument in which we build up the product in (*) by repeated multiplication on the right by terms α_{p^i} .

To prove $(**)$ we recall that $\alpha_n \alpha_{p^k} = \binom{n+p^k}{n} \alpha_{n+p^k}$. The mod p value of this binomial coefficient can be computed using Lemma 3C.6 below. Assuming that $n_i < p$ for each i and that $n_k + 1 < p$, the p -adic representations of $n + p^k$ and n differ only in the coefficient of p^k , so mod p we have $\binom{n+p^k}{n} = \binom{n_k+1}{n_k} = n_k + 1$. This conclusion also holds if $n_k + 1 = p$, when the p -adic representations of $n + p^k$ and n differ also in the coefficient of p^{k+1} . The statement $(**)$ then follows.

Lemma 3C.6. *If p is a prime, then $\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p}$ where $n = \sum_i n_i p^i$ and $k = \sum_i k_i p^i$ with $0 \leq n_i < p$ and $0 \leq k_i < p$ are the p -adic representations of n and k .*

Here the convention is that $\binom{n}{k} = 0$ if $n < k$, and $\binom{n}{0} = 1$ for all $n \geq 0$.

Proof: In $\mathbb{Z}_p[x]$ there is an identity $(1+x)^p = 1+x^p$ since p clearly divides $\binom{p}{k} = p!/k!(p-k)!$ for $0 < k < p$. By induction it follows that $(1+x)^{p^i} = 1+x^{p^i}$. Hence if $n = \sum_i n_i p^i$ is the p -adic representation of n then:

$$\begin{aligned} (1+x)^n &= (1+x)^{n_0} (1+x^p)^{n_1} (1+x^{p^2})^{n_2} \cdots \\ &= \left[1 + \binom{n_0}{1}x + \binom{n_0}{2}x^2 + \cdots + \binom{n_0}{p-1}x^{p-1} \right] \\ &\quad \times \left[1 + \binom{n_1}{1}x^p + \binom{n_1}{2}x^{2p} + \cdots + \binom{n_1}{p-1}x^{(p-1)p} \right] \\ &\quad \times \left[1 + \binom{n_2}{1}x^{p^2} + \binom{n_2}{2}x^{2p^2} + \cdots + \binom{n_2}{p-1}x^{(p-1)p^2} \right] \times \cdots \end{aligned}$$

When this is multiplied out, one sees that no terms combine, and the coefficient of x^k is just $\prod_i \binom{n_i}{k_i}$ where $k = \sum_i k_i p^i$ is the p -adic representation of k . $\square$

Pontryagin Product

Another special feature of H-spaces is that their homology groups have a product operation, called the **Pontryagin product**. For an H-space X with multiplication $\mu: X \times X \rightarrow X$, this is the composition

$$H_*(X; R) \otimes H_*(X; R) \xrightarrow{\times} H_*(X \times X; R) \xrightarrow{\mu_*} H_*(X; R)$$

where the first map is the cross product defined in §3.B. Thus the Pontryagin product consists of bilinear maps $H_i(X; R) \times H_j(X; R) \rightarrow H_{i+j}(X; R)$. Unlike cup product, the Pontryagin product is not in general associative unless the multiplication μ is associative or at least associative up to homotopy, in the sense that the maps $X \times X \times X \rightarrow X$, $(x, y, z) \mapsto \mu(x, \mu(y, z))$ and $(x, y, z) \mapsto \mu(\mu(x, y), z)$ are homotopic. Fortunately most H-spaces one meets in practice satisfy this associativity property. Nor is the Pontryagin product generally commutative, even in the graded sense, unless μ is commutative or homotopy-commutative, which is relatively rare for H-spaces. We will give examples shortly where the Pontryagin product is not commutative.

In case X is a CW complex and μ is a cellular map the Pontryagin product can be computed using cellular homology via the cellular chain map

$$C_i(X; R) \times C_j(X; R) \xrightarrow{\times} C_{i+j}(X \times X; R) \xrightarrow{\mu_*} C_{i+j}(X; R)$$

where the cross product map sends generators corresponding to cells e^i and e^j to the generator corresponding to the product cell $e^i \times e^j$, and then μ_* is applied to this product cell.

Example 3C.7. Let us compute the Pontryagin product for $J(S^n)$. Here there is one cell e^{in} for each $i \geq 0$, and μ takes the product cell $e^{in} \times e^{jn}$ homeomorphically onto the cell $e^{(i+j)n}$. This means that $H_*(J(S^n); \mathbb{Z})$ is simply the polynomial ring $\mathbb{Z}[x]$ on an n -dimensional generator x . This holds for n odd as well as for n even, so the Pontryagin product need not satisfy the same general commutativity relation as cup product. In this example the Pontryagin product structure is simpler than the cup product structure, though for some H-spaces it is the other way round. In applications it is often convenient to have the choice of which product structure to use.

This calculation immediately generalizes to $J(X)$ where X is any connected CW complex whose cellular boundary maps are all trivial. The cellular boundary maps in the product X^m of m copies of X are then trivial by induction on m using Proposition 3B.1, and therefore the cellular boundary maps in $J(X)$ are all trivial since the quotient map $X^m \rightarrow J_m(X)$ is cellular and each cell of $J_m(X)$ is the homeomorphic image of a cell of X^m . Thus $H_*(J(X); \mathbb{Z})$ is free with additive basis the products $e^{n_1} \times \cdots \times e^{n_k}$ of positive-dimensional cells of X , and the multiplicative structure is that of polynomials in noncommuting variables corresponding to the positive-dimensional cells of X .

Another way to describe $H_*(J(X); \mathbb{Z})$ in this example is as the tensor algebra $T\tilde{H}_*(X; \mathbb{Z})$, where for a graded R -module M that is trivial in dimension zero, like the reduced homology of a path-connected space, the tensor algebra TM is the direct sum of the n -fold tensor products of M with itself for all $n \geq 1$, together with a copy of R in dimension zero, with the obvious multiplication coming from tensor product and scalar multiplication.

Generalizing the preceding example, we have:

Proposition 3C.8. *If X is a connected CW complex with $H_*(X; R)$ a free R -module, then $H_*(J(X); R)$ is isomorphic to the tensor algebra $T\tilde{H}_*(X; R)$.*

This can be paraphrased as saying that the homology of the free H-space generated by a space with free homology is the free algebra generated by the homology of the space.

Proof: With coefficients in R , let $\varphi: T\tilde{H}_*(X) \rightarrow H_*(J(X))$ be the homomorphism whose restriction to the n -fold tensor product $\tilde{H}_*(X)^{\otimes n}$ is the composition

$$\tilde{H}_*(X)^{\otimes n} \hookrightarrow H_*(X)^{\otimes n} \xrightarrow{\times} H_*(X^n) \rightarrow H_*(J_n(X)) \rightarrow H_*(J(X))$$